

# Informe de experimento: react\_nogate

Generado el 2026-09-16 13:59 por benchmarks/experiment\_report.py a partir de 1 report(s) del directorio react\_nogate (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

## 1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: react\_nogate
- Casos: case14
- Requests: generadas, N=40 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=10, parameterized=10, multistep=10, ambiguous=10
- runs por item: 1; k (perturbación  $\pm 10\% \cdot k$ ): 1; seeds: [0]; max\_rounds: 8; temperature: 0.0; timeout\_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 13:31
- Costo total: 1.3219 USD (40 items; costo medio 0.03305 USD/item)
- Items totales: 40; ok=40; con error=0; tokens totales=458729
- Solved global: 62.5 % (25/40)

Errores por tipo:

(ningún item con error)

## 2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt\_variants.build\_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

### Método react\_nogate

**System prompt (SYSTEM\_PROMPT\_EN)** —1375 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

**User prompt (primer item)** = request\_text —67 caracteres; case=case14, seed=1911784258, k=1, request\_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

### 3. Resultados agregados

Columnas del protocolo (medias por método  $\times$  caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfactagent_definicion` **utilidad de tarea** Form. (formulación exacta de las tool calls), V\_MAE (p.u.) y F\_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B\_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

#### Por método $\times$ caso

| Method       | Case   | N  | Form.        | V_MAE    | F_MAE | Solved       | Solver status | B_mean | Faith. | SFR | Calls | Tok.   | Claim      | Abst.        | V pass | \$ |
|--------------|--------|----|--------------|----------|-------|--------------|---------------|--------|--------|-----|-------|--------|------------|--------------|--------|----|
| react_nogate | case14 | 40 | 62.5 (25/40) | 1.13e-04 | 0.01  | 62.5 (25/40) | 100.0 (40/40) | 0      | 99.0   | -   | 2.8   | 11468- | 0.0 (0/40) | 90.0 (36/40) | 1.3219 |    |

#### Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

| Method       | plain       | parameterized | multistep   | ambiguous   |
|--------------|-------------|---------------|-------------|-------------|
| react_nogate | 80.0 (8/10) | 90.0 (9/10)   | 40.0 (4/10) | 40.0 (4/10) |

#### Solved por dificultad (% , items resueltos / items)

| Method       | plain       | parameterized | multistep   | ambiguous   |
|--------------|-------------|---------------|-------------|-------------|
| react_nogate | 80.0 (8/10) | 90.0 (9/10)   | 40.0 (4/10) | 40.0 (4/10) |

#### Tipos de error de formulación (conteo de items)

| Method       | extra_arg_changes | result ok |
|--------------|-------------------|-----------|
| react_nogate | 15                | 25        |

#### Errores de ejecución por método (conteo de items con error)

(ningún item con error)

### 4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed\_success\_on\_failure, stale\_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

#### Método react\_nogate

##### Ejemplo: resuelto (solved=True)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
  ↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 5930 chars más}
  gate: PASS (reasons=-, mismatch=7.85e-09 MW)
--- Ronda 3 ---
LLM: 1. One-sentence summary
```

```

I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the post-outage power flow converged.

2. Key numbers
- Total load: 253.846 MW
- Total generation: 266.510 MW
- Total losses: 12.665 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if ...[504 chars más]
--- Respuesta final ---
1. One-sentence summary
I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the post-outage power flow converged.

2. Key numbers
- Total load: 253.846 MW
- Total generation: 266.510 MW
- Total losses: 12.665 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if any
- Voltage violations:
  - Bus 1: 1.060 pu, overvoltage
  - Bus 6: 1.070 pu, overvoltage
  - Bus 7: 1.057 pu, overvoltage
  - Bus 8: 1.090 pu, overvoltage
  - Bus 11: 1.064 pu, overvoltage
  - Bus 12: 1.054 pu, overvoltage
- Thermal violations: none
- Disconnected line status:
  - Line 10-11 flow is 0.0 MW / 0.0 MVar as expected

4. Suggested next step
If you want, I can now check the most loaded branch, run an N-1 contingency analysis, or suggest remedial actions for the voltage violati ...
↪ [4 chars más]
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=5350;
↪ cost_usd=0.0173; wall=7.0 s

```

## Ejemplo: formulación fallida (formulation\_exact=False)

```

request(case14-ambiguous-011-s0, case14, difficulty=ambiguous, seed=1971893210): Load case14 and set the active load at bus 6 to 14,700 kw.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 6, "p_mw": 14.7})
executed_calls: load_case({"case_name": "case14"}) -> get_status({}) -> modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 11.2})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 263.13580543670497,
↪ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: get_status({})
output: {"active_case": "case14", "network_info": {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11,
↪ "total_load_mw": 263.13580543670497, "total_gen_capacity_mw": 772.4000000000001, "has_last_result": false}
gate: n/a (no es un resultado de flujo)
--- Ronda 3 ---
tool: modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 11.2})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.236323374534588, "is_violation": false,
↪ "violation_type": null}, {"bus_id": 3, "vm_p ...[6297 chars más]
gate: PASS (reasons=-, mismatch=6.47e-09 MW)
--- Ronda 4 ---
LLM: 1. Summary: I loaded IEEE case14 and set bus 6 active load to 14,700 kW (interpreted as 14.7 MW), and the power flow converged.

2. Key numbers:
- Total load: 267.145 MW
- Total generation: 281.819 MW
- Total losses: 14.674 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if any:
- Voltage v ...[257 chars más]
--- Respuesta final ---
1. Summary: I loaded IEEE case14 and set bus 6 active load to 14,700 kW (interpreted as 14.7 MW), and the power flow converged.

2. Key numbers:
- Total load: 267.145 MW
- Total generation: 281.819 MW
- Total losses: 14.674 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if any:
- Voltage violations: overvoltage at buses 1, 6, 7, 8, 9, 10, 11, and 12
- Thermal violations: none

4. Suggested next step:
- If you want, I can also adjust the reactive load at bus 6, check the most loaded branch, or run a contingency analysis on this modified
↪ case.

```



- $V\_MAE \geq 1e-4$  p.u. en 1 filas método×caso con solver: react\_nogate/case14 (1.13e-04).
- El gate verificó 58 resultados de flujo, 0 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved de react\_nogate: 62.5 % (25/40).
- Formulación exacta agregada sobre métodos con tools: 62.5 % (25/40).
- Formulación por dificultad: mejor parameterized 90.0 % (9/10), peor multistep 40.0 % (4/10).
- Errores de formulación más frecuentes: extra\_arg\_changes\_result=15.
- Ningún item reclamó éxito sobre un fallo (claimed\_success\_on\_failure=0).
- 1 de 26 items con mutación de red reportaron estado obsoleto (stale\_state).
- Trazabilidad de números (Faith.) media con tools: 99.0 %.
- Costo total 1.3219 USD; el método más caro fue react\_nogate con 1.3219 USD (100 %).
- Métodos con tools: 2.83 tool calls medias por item (máximo 6).

## 6. Archivos

Rutas absolutas bajo react\_nogate.report.rescored.json (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/report.json [leído:
↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.rescored.json
scoreboard_per_case.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard_per_case.rescored.json
traces (40 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/traces
```